Political Risk and Exchange Rates: Korea's Martial Law Declaration

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We estimate the effect of political risk on exchange rates by exploiting a series of political risk events following the martial law declaration in Korea on December 3, 2024. The exchange rates for 37 countries are decomposed into a global dollar factor and country-specific local factors using a dynamic factor model. The local factors are then used in an event-study difference-in-differences regression with Korea as the treated unit. We find 2.35 percentage points of Korea-specific excess depreciation in December 2024, 73% of the month's total depreciation of 3.20 percentage points. A further investigation with daily data shows that this excess depreciation tracks the political events of the month. By the close of the declaration day, the excess depreciation already stood at three quarters of its December average in the daily data, the largest move among the 37 currencies that day. The withdrawal of martial law reversed the move only partly, and the level rose again during the impeachment process.

Keywords: Exchange Rate, Political Risk, Dynamic Factor Model, Difference in Differences, Randomization Inference, Korean Won, Martial Law

JEL codes: F31, F50

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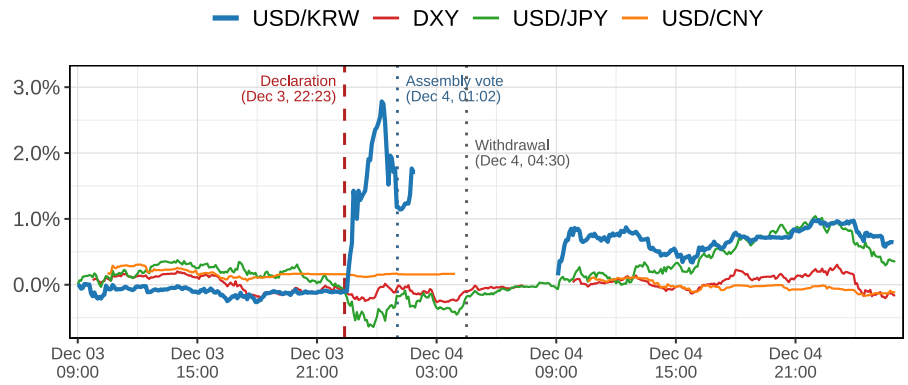
1. Introduction

On December 3, 2024, the President of South Korea announced martial law at 22:23. It was an unanticipated late-night shock to the foreign exchange (FX) market. The declaration lasted only about six hours. The National Assembly voted 190–0 to demand its lifting, and the President withdrew it early the next morning. Nevertheless, the US dollar–Korean won exchange rate (USD/KRW) spiked immediately at the announcement (i.e., the Korean won depreciated). It did not return to its pre-announcement level (Figure 1A). Political events continued until the Constitutional Court upheld the impeachment on April 4, 2025. We call this period, from the declaration through the Court ruling, the political risk episode (see the timeline in Appendix A).

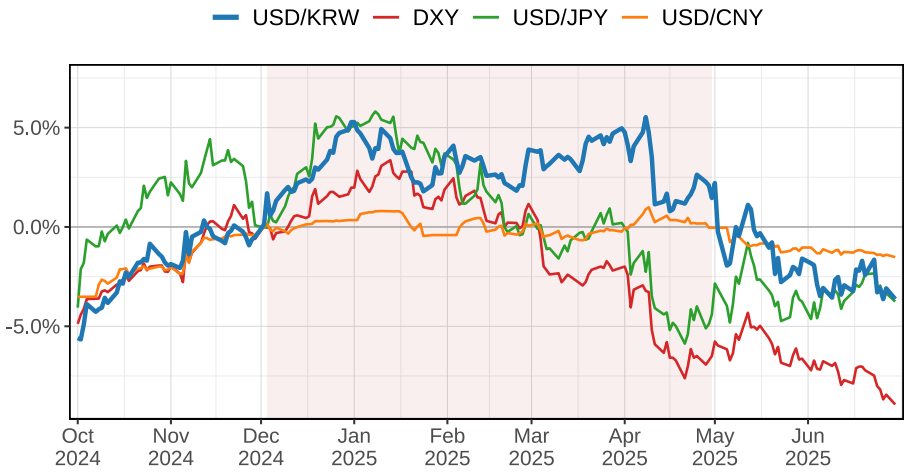
How much of this depreciation reflects political risk? One could follow recent studies (Bossman, Gubareva, and Teplova 2023; Hossain, Masum, and Saadi 2024; Liu and Zhang 2024; Yilmazkuday 2025) and address this question using a news-based index of geopolitical risk. Since our question concerns the depreciation associated with a specific domestic political shock, we instead exploit the timing of a specific political event, as in Dräger, Gründler, and Potrafke (2025), Karoubi and Seeballack (2026), and Aizenman and Saadaoui (2026). The closest work is Kim (2026), who examines the same martial law declaration and finds a within-country flight to quality in Korean equities that dissipates within days. We complement this work by measuring the won's excess depreciation and its response to subsequent political events.

Isolating the effect of political risk on exchange rate movements faces five challenges. First, the exchange rate is a relative price: USD/KRW may rise not only due to KRW weakness but also due to broad USD strength that simultaneously affects all currencies. Second, even after removing broad USD strength, there could have been concurrent shocks common to a group of countries including Korea. As Figure 1B shows, the Japanese yen (JPY) and the Chinese yuan (CNY) depreciated alongside KRW from December 2024. A simple before and after comparison of KRW would therefore overstate the political risk contribution. Third, US tariff announcements by the Trump administration coincided with the tail end of Korea's political risk episode. Because Korea has high trade exposure and is sensitive to US trade policy, these tariff threats could otherwise contaminate the estimated political risk effect. Fourth, the martial law declaration must have been unanticipated. If markets had priced in the political risk before December 2024, the estimate would understate the true impact. Fifth, the comparison itself must remain clean: Korea's shock must not spill over to the control currencies or enter the estimated dollar factor.

To address these challenges, we employ a two-stage empirical strategy. First, we use a dynamic factor model (DFM) to extract a global dollar factor from the 36 control exchange



A. Intraday: December 3–4, 2024



B. Daily: October 2024 – June 2025

FIGURE 1. Exchange Rates Around Korea's Martial Law Declaration

Notes: Both panels show exchange rates normalized to zero at December 3, 2024, 09:00 KST. Panel B plots daily closes relative to that quote. For exchange rates, an increase denotes depreciation against the US dollar. For the U.S. Dollar Index (DXY), it denotes broad dollar appreciation. Panel A plots Bloomberg 5-minute intraday data. The quotes are indicative composite prices, not executable trades. Executed onshore trades track them closely through the event window (Figure B.2). Gaps are market closures. All times are Korea Standard Time (KST = UTC+9). Vertical lines mark the martial law announcement (dashed red, 22:23, December 3) the National Assembly vote demanding the lifting of martial law (dotted blue, 01:02, December 4), and the withdrawal of martial law (dotted gray, 04:30, December 4). Panel B shades the political risk period (December 2024–April 2025).

rates, addressing the first challenge.[1] We then use the local factor—the residual exchange rates with the global dollar factor removed—in an event-study difference-in-differences (DiD). In the DiD regression with November 2024 as the reference period, Korea is the treated unit and 36 never-treated countries are controls. The time fixed effects in the DiD absorb any remaining shocks common to the full panel that are not captured by the estimated global factor, while the broad control group limits the influence of region-specific comovement. This setup allows our estimate to reflect Korea's excess depreciation beyond what the control countries experienced, addressing the second challenge. A Korea-specific shock unrelated to politics could still coincide with the episode. We therefore apply the same event-study design to daily data to examine the won's response to individual political events. The third challenge is addressed by the timing of the event. The December estimate predates the tariff shock, which began with the new US administration in 2025. We nevertheless verify that adjusting the outcome for each currency's trade policy sensitivity leaves the December estimate essentially unchanged (Appendix D.3). The fourth challenge is addressed by the evidence showing no visible anticipation of the declaration (Appendix B). The fifth challenge is addressed by the design of the first stage and by the breadth of the control group. The factor is estimated on the 36 control currencies alone over the pre-event sample. Hence, no Korean data enter the dollar factor, and no post-event information enters its parameters. Spillovers that move the control currencies in the same direction as the won would bias the estimate toward zero.

We find 2.35 percentage points of Korea-specific excess depreciation in December 2024. A model-implied decomposition shows that this accounts for 73% of the total monthly depreciation. Because Korea is the only treated country, our statistical conclusions rest on randomization inference. Korea's December estimate is the third largest in absolute value among all 37 possible treatment assignments with a placebo p-value of 0.081. We then ask, using daily data, which political events are related to this excess depreciation. By the close of December 3 the excess depreciation already stood at three quarters of its December average, the largest move among the 37 currencies that day. The withdrawal of martial law the next morning unwound only part of it. The level rose again over the impeachment process that followed.

[1] The factor model is widely used in the international finance literature to analyze exchange rates. See, for example, Lustig, Roussanov, and Verdelhan (2011) and Verdelhan (2018).

2. Empirical Methodology and Data

2.1. Factor model

We employ a DFM as below. For each currency of country i at time t,

$$\Delta s_{it} = \mu_i + \lambda_i^G f_t^G + e_{it}, \tag{1}$$

where Δs_{it} is the monthly log change in the USD/currency-i exchange rate, so that $\Delta s_{it} > 0$ means depreciation against the US dollar. Here μ_i is the currency-specific mean change, f_t^G is the global dollar factor, λ_i^G is its loading, and e_{it} is the country-specific residual after removing the mean and the global dollar factor. The factor follows an AR(1) process, $f_t^G = \phi\, f_{t-1}^G + \eta_t$.

We estimate the model so that no post-event information and no Korean data enter the estimation of the factor model. In particular, the parameters are estimated on the 36 control currencies excluding Korea over the pre-event sample (February 2000 through November 2024). Korea's loading is estimated by projecting KRW changes on the factor over the same pre-event sample. With the estimated parameters held fixed, the factor over the event period is obtained from the one-sided Kalman filter. The country-specific local factor is defined as the residual $\hat{e}_{it}$ computed as $\hat{e}_{it} = \Delta s_{it} - \hat{\mu}_i - \hat{\lambda}_i^G \hat{f}_t^G$, where $\hat{\mu}_i$ is the pre-event mean.

2.2. Difference in differences

We then apply an event-study DiD to estimate the effect of political risk on USD/KRW. In particular, we use the country-specific local factor as the outcome variable of the DiD regression. Specifically, we cumulate $\hat{e}_{it}$ over time and recenter at November 2024. The outcome then measures each currency's cumulative depreciation beyond what global dollar movements would predict, relative to the level before the shock. Formally, $\tilde{E}_{it} = \sum_{s \le t} \hat{e}_{is} - \sum_{s \le t^*} \hat{e}_{is}$ where t^* denotes November 2024. The treatment group is Korea and the control group consists of all 36 never-treated countries. The treatment is the onset of the political risk episode in December 2024 ($k = 0$). The regression panel runs from January 2024 through December 2025 (888 country month observations = 37 countries × 24 months).

The main regression is

$$\tilde{E}_{it} = \alpha_i + \gamma_t + \sum_{k \in \mathcal{K}} \beta_k \cdot (\text{Korea}_i \times \mathbf{1}[t = k]) + \varepsilon_{it}, \tag{2}$$

where α_i and γ_t are country and time fixed effects, Korea_i is an indicator for the treatment group, and $\mathbf{1}[t = k]$ is an indicator for event month k relative to December 2024, with

November 2024 ($k = -1$) as the omitted reference month. Our specification is saturated: every panel month except the November 2024 reference receives its own Korea indicator. Hence $\mathcal{K} = \{-11, \ldots, -2, 0, \ldots, 12\}$ covers January 2024 through December 2025.

Our statistical conclusions rest on randomization inference (Appendix F), while we report clustered standard errors for descriptive purposes. The coefficient β_k measures Korea-specific excess depreciation relative to the control currencies in event month k, after controlling for country and time fixed effects.[2]

2.3. Data

We use monthly exchange rates for the 37 currencies against the US dollar, covering countries spanning Asia and Pacific, Europe, the Americas, and Africa. The DiD panel covers January 2024 through December 2025. Table 1 focuses on the political risk episode (December 2024–April 2025) and the pre-trend month (October 2024), relative to the reference month (November 2024). The complete set of coefficients is in Appendix E, and data sources and the full country list are described in Appendix C.

We also use daily frequency data to examine the daily movements around individual political events in December (Section 3.2).[3] The daily data cover the same countries over January 2021 through April 2025.

3. Results

3.1. Event study estimates

Figure 2 presents the estimated coefficients $\hat{\beta}_k$ for Korea (in red) together with the same coefficient path for each of the 36 possible placebo assignments (in gray). Korea lies at the upper edge of the randomization distribution spanned by the placebos, most clearly in December. After December, while Korea's path stays elevated, the placebo distribution is more dispersed. Table 1 reports the underlying regression estimates.[4]

The results show a large Korea-specific excess depreciation at the onset of the political risk episode. In December 2024, KRW depreciated by 2.35 percentage points beyond the control country average. To gauge the magnitude of this estimate, monthly USD/KRW log changes have a standard deviation of about 2.2 pp over 2000–2025. The December 2024 estimate is about one standard deviation of the historical monthly variation. Identified

[2] Combining equations (1) and (2), the two-stage setup has an interactive fixed effects interpretation (Bai 2009). Appendix E gives the representation.

[3] We thank the Editor for this suggestion. See Appendices H and E.2 for more details.

[4] The pre-treatment coefficients are consistent with parallel trends. Appendix E.3 reports the complete pre-treatment paths and the joint placebo tests.

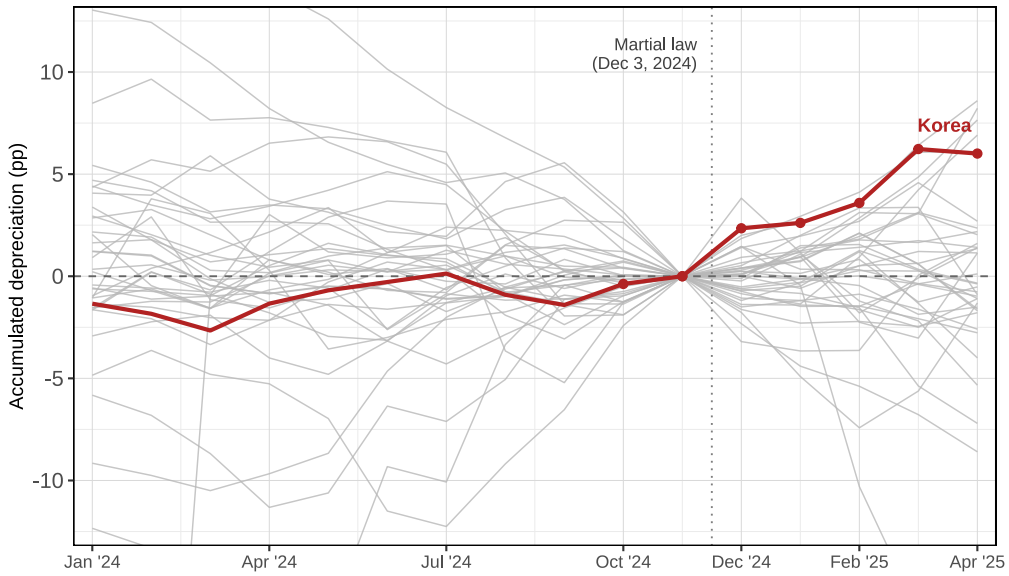


FIGURE 2. Effect of Korea's Political Risk on USD/KRW

Notes: The red line is Korea's event-study coefficient path from equation (2). The outcome is the accumulated local factor $\tilde{E}_{it}$ (log level, re-centered at November 2024 = 0). Each gray line repeats the entire two-stage estimation with one of the 36 control currencies designated treated. The figure shows January 2024 through April 2025, and five placebo paths exceed the displayed range. The full window through December 2025 is in Appendix F (Figure F.2).

TABLE 1. Effect of Korea's Political Risk on USD/KRW

	Estimate (pp)	Rank of 37	Placebo p
Oct 2024 ($k = -2$)	−0.38	28	0.757
	(0.22)		
Dec 2024 ($k = 0$)	2.35	3	0.081
	(0.22)		
Jan 2025 ($k = 1$)	2.61	6	0.162
	(0.30)		
Feb 2025 ($k = 2$)	3.59	6	0.162
	(0.50)		
Mar 2025 ($k = 3$)	6.23	4	0.108
	(0.69)		
Apr 2025 ($k = 4$)	6.01	9	0.243
	(0.83)		

Notes: Dependent variable is the cumulative local factor (pp), re-centered at November 2024, in the saturated event study of equation (2) with country and time fixed effects. Clustered standard errors in parentheses are descriptive. Rank is the position of Korea's coefficient among the 37 assignments in absolute value, and the placebo p-value is two-sided and inclusive (Appendix F). The full set of 23 coefficients is in Appendix E.

estimates for other major political events in the literature range from about 0.35% to 7%.[5]

Korea's December coefficient is the third largest of the 37 possible treatment assignments in absolute value with a two-sided placebo p-value of 0.081. The two larger placebo estimates belong to Brazil (3.82 pp, rank 1) and Israel (−3.20 pp, rank 2), both country-specific episodes unrelated to Korea (Appendix F).

The point estimates did not reverse. The January 2025 estimate is 2.61 pp, February is 3.59 pp, and March rises further to 6.23 pp. In April, at the end of the episode, the estimate edges down to 6.01 pp. However, none of the coefficients after December 2024 is distinguishable from zero under the randomization inference, with placebo p-values of 0.108 or larger (Appendix G). The later estimates therefore are too imprecise to distinguish persistence, decay, or disappearance.

3.2. Decomposition of the December depreciation

Monthly. The December estimate has an exact model-implied accounting interpretation. Evaluating equation (1) for Korea in December 2024 and subtracting the control average of the local changes,

$$\underbrace{\Delta s_{KRW,\text{Dec}}}_{\text{observed: 3.20}} = \underbrace{\hat{\mu}_{KRW}}_{\text{mean: 0.07}} + \underbrace{\hat{\lambda}^{G}_{KRW}\hat{f}^{G}_{\text{Dec}}}_{\text{dollar factor: 0.89}} + \underbrace{\bar{\hat{e}}_{-KRW,\text{Dec}}}_{\text{controls' local: }-0.10} + \underbrace{\hat{\beta}_0}_{\text{Korea-specific excess: 2.35}}, \quad (3)$$

where $\bar{\hat{e}}_{-KRW,t}$ is the average local change of the 36 controls, and $\hat{\beta}_0 = \hat{e}_{KRW,t} - \bar{\hat{e}}_{-KRW,t}$ is the December coefficient. The four components sum to the observed 3.20 pp up to rounding. The Korea-specific excess component, 2.35 pp, accounts for roughly 73% of the observed depreciation that month.[6]

Daily. At daily frequency, we examine which political events are related to the Korea-specific excess depreciation identified at the monthly frequency. The daily model is estimated separately and uses the December 2 close as its reference, so its coefficients describe the within-month path and need not reproduce the monthly estimate. Let $\hat{\beta}^{d}_{s}$ denote the estimated daily event-study coefficient for weekday s, the analogue of $\hat{\beta}_k$ with the December 2 close as the reference.[7] The one-day excess move is $\Delta\hat{\beta}^{d}_{s} = \hat{\beta}^{d}_{s} - \hat{\beta}^{d}_{s-1}$. The

[5] See Balima (2020) for coup d'état estimates and Manasse, Moramarco, and Trigilia (2024) for Brexit referendum estimates. We present this comparison as context rather than as validation of our estimate. We thank an anonymous referee for suggesting this comparison.

[6] The 73% is a descriptive model-implied ratio. Across the reported specifications it ranges from 66% to 74% (Table 3). Being a fixed rescaling of the December estimate, its statistical uncertainty is that of the numerator (Table E.1).

[7] Appendix E.2 states the daily regression. Appendix H describes the daily factor.

end-of-December level is decomposed as:

$$\underbrace{\hat{\beta}^{d}_{\text{Dec 31}}}_{\text{end of December: 3.93}} = \underbrace{\Delta\hat{\beta}^{d}_{\text{Dec 3}}}_{\text{declaration: 1.67}} + \underbrace{\Delta\hat{\beta}^{d}_{\text{Dec 4}}}_{\text{withdrawal: } -1.06} + \cdots + \underbrace{\Delta\hat{\beta}^{d}_{\text{Dec 26}}}_{\text{court dispute: 0.69}} + \cdots. \quad (4)$$

Table 2 reports these one-day moves for the major political events in December, together with their ranks and placebo p-values. The declaration day produced a 1.67 pp excess depreciation, the largest move in absolute value among the 37 treatment assignments that day (rank 1, $p = 0.027$).[8] By the close of December 3 the level already stood at 75% of the daily December average of 2.22 pp. The withdrawal of martial law on December 4 brought a 1.06 pp reversal, again the largest absolute move among the 37 assignments that day (rank 1, $p = 0.027$). The dispute over the Constitutional Court appointments on December 26 ranks third ($p = 0.081$).[9] The other political event days show no unusual one-day move, with ranks between 7 and 36. The moves cumulate to $\hat{\beta}^{d}_{\text{Dec 31}} = 3.93$ pp by the end of December (rank 2, $p = 0.054$), so the level rose again over the impeachment process and ended the month above its December 3 close. Table E.3 reports the full path.

TABLE 2. One-Day Moves on the December Event Days

	$\Delta\hat{\beta}^{d}_{s}$ (pp)	Rank of 37	Placebo p
December 3, declaration	1.67	1	0.027
December 4, withdrawal	-1.06	1	0.027
December 9, after the failed impeachment vote (Dec 7)	0.66	7	0.189
December 12, president's address	-0.06	24	0.649
December 16, after the impeachment vote (Dec 14)	0.18	16	0.432
December 26, Constitutional Court appointments dispute	0.69	3	0.081
December 27, acting president impeached	0.01	36	0.973
December 31, arrest warrant	0.23	11	0.297
Level at the December 31 close, $\hat{\beta}^{d}_{\text{Dec 31}}$	3.93	2	0.054

Notes: One-day moves $\Delta\hat{\beta}^{d}_{s} = \hat{\beta}^{d}_{s} - \hat{\beta}^{d}_{s-1}$ of the daily event-study coefficient between consecutive weekdays, i.e., of Korea's excess depreciation on the December event days, with weekend events assigned to the following Monday. The last row is the level at the December 31 close, the sum of the daily moves after the December 2 reference close. The construction is described in Appendix E.2. Rank and inclusive two-sided placebo p-value from re-estimating the full procedure for each of the 37 possible treatment assignments.

[8] The KRW daily close is the 02:00 KST close of the Seoul extended session, so the December 3 return runs to 02:00 on December 4 and contains both the declaration and the Assembly vote.

[9] With three of the Court's nine seats vacant, upholding the impeachment required all six sitting justices, so the decision not to fill them raised uncertainty over the outcome.

4. Robustness

Table 3 collects the December 2024 estimate across every specification for robustness analysis: alternative factor constructions, alternative control groups, and adjustments for trade policy uncertainty. Across the event-study variants the estimate lies between 2.12 and 2.37 pp, and the share of the December depreciation between 66% and 74%. The point estimates are similar across designs, while the placebo p-values range from 0.081 to 0.135. Appendix D reports each variant in full.

TABLE 3. December 2024 Estimate Across Designs

Design	December (pp)	Share (%)	Randomization inference
Baseline	2.35	73	rank 3 of 37, $p = 0.081$
Alternative factor constructions			
full-sample smoother, Korea included	2.30	72	rank 3 of 37, $p = 0.081$
pre-event factor, Korea included	2.34	73	rank 3 of 37, $p = 0.081$
principal component	2.31	72	rank 4 of 37, $p = 0.108$
two-factor model	2.37	74	rank 3 of 37, $p = 0.081$
three-factor model	2.36	74	rank 3 of 37, $p = 0.081$
floating factor universe[a]	2.32	72	rank 3 of 27, $p = 0.111$
Alternative control groups			
floating controls[a]	2.34	73	rank 3 of 27, $p = 0.111$
best pre-event factor fit[b]	2.12	66	rank 4 of 37, $p = 0.108$
Trade policy adjustments			
TPU-adjusted outcome (three variants)	2.24–2.33	70–73	rank 5 of 37, $p = 0.135$

Notes: Each row re-estimates the December 2024 effect under one design change, holding the saturated event study fixed unless noted. Share is the December estimate divided by the observed December depreciation of 3.20 pp, a descriptive model-implied ratio. Randomization inference re-runs the full two-stage procedure, including the row's own factor construction and control group, for every possible treatment assignment. Details in Appendix D. [a] These rows restrict the currencies to the floating group, so the rank is among the 27 floating assignments: Korea and the 26 floating controls of Table C.2. [b] This row restricts the controls to the 18 currencies with the best pre-event factor fit (Appendix D). The three TPU variants (level, change, and innovation of the index) all give rank 5 of 37 (Table D.4).

5. Conclusion

We estimate the effect of political risk on exchange rates during Korea's political risk episode. The won shows 2.35 percentage points of Korea-specific excess depreciation in December 2024, 73% of the month's depreciation. Daily data tie it to December's political events. In the daily data, three quarters of the December average was in place by the close of the declaration day. The withdrawal reversed only part of it, and the level rose again over the impeachment process.

We build on recent evidence on geopolitical risk in currency markets (Yilmazkuday 2025) by measuring Korea-specific excess depreciation during a domestic political risk

episode. The daily analysis relates this depreciation to individual political events. We also complement Kim's (2026) evidence of a temporary within-country flight to quality in equities: the won's excess depreciation did not fully reverse within December.

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Appendix A. Timeline of Key Events

Table A.1 lists the key events from December 2024 to April 2025: the martial law declaration, the impeachment proceedings, and the arrests.

TABLE A.1. Timeline of Korea's Political Risk Episode, December 2024–April 2025

Date	Event
Dec 3, 2024	President Yoon Suk-yeol declares martial law in an unscheduled televised address beginning at 22:23 KST. The declaration takes formal effect at 22:27, citing "anti-state forces" in the opposition. Martial Law Proclamation No. 1 places all media under martial law command. Troops are deployed to the National Election Commission from ~22:30 and reach the National Assembly after midnight.
Dec 4, 2024	The National Assembly votes 190–0 to demand the lifting of martial law at 01:02. Martial law is lifted at 04:30, about six hours after the declaration.
Dec 7, 2024	First impeachment motion fails. The ruling People Power Party (PPP) walks out, quorum not met.
Dec 12, 2024	Yoon's televised address: he defends the declaration as an act of governance and says he will "fight to the end."
Dec 14, 2024	Second impeachment passes 204–85, requiring at least 12 ruling-party votes in the secret ballot. Yoon suspended. Prime Minister Han Duck-soo becomes acting president. Constitutional Court review begins.
Dec 26, 2024	Acting President Han declines to appoint three Constitutional Court justices. The opposition files a motion to impeach him.
Dec 27, 2024	Acting President Han Duck-soo is impeached. Finance Minister Choi Sang-mok becomes acting president.
Dec 31, 2024	The Seoul Western District Court issues an arrest warrant for Yoon, the first for a sitting president. Acting President Choi appoints two Constitutional Court justices.
Jan 15, 2025	Yoon arrested by the Corruption Investigation Office for High-ranking Officials (CIO)—the first arrest of a sitting Korean president.
Jan 26, 2025	Yoon formally indicted on insurrection charges.
Feb–Mar 2025	Constitutional Court holds oral arguments.
Apr 4, 2025	Constitutional Court upholds impeachment 8–0, permanently removing Yoon from office. KRW partially recovers.

Appendix B. Evidence of No Visible Anticipation

The evidence below shows no visible anticipation of Korea's martial law declaration on December 3, 2024 by the markets or the general public. It also confirms that the immediate after-hours reaction consists of executed onshore trades.

NDF 1-month rate. Figure B.1 plots the USD/KRW 1-month non-deliverable forward (NDF) rate at intraday frequency from December 2 through December 6, 2024. The NDF market is the primary channel through which offshore investors hedge or take speculative positions on USD/KRW. The NDF shows no run-up before the announcement and moves sharply immediately after 22:23 on December 3.

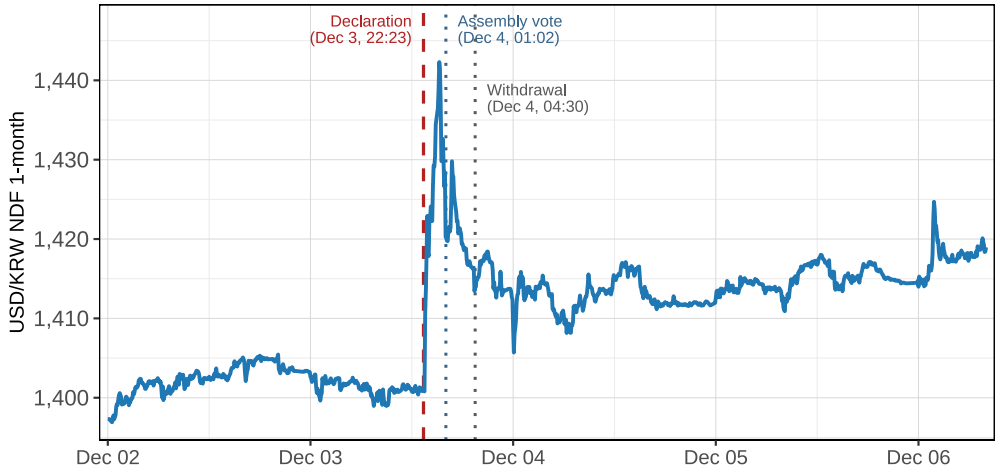


FIGURE B.1. USD/KRW NDF 1-Month Rate Around the Martial Law Declaration

Notes: USD/KRW 1-month NDF, Bloomberg 5-minute intraday data, December 2–6, 2024 (KST). The dashed red vertical line marks the martial law announcement (22:23, December 3, 2024). The dotted blue vertical line marks the National Assembly vote demanding the lifting of martial law (01:02, December 4, 2024). The dotted gray vertical line marks the withdrawal of martial law (04:30, December 4, 2024).

Onshore trading prices. The after-hours movement in Figure 1A is confirmed by trading prices. The Seoul interbank market's extended session runs to 02:00, so onshore trading was open at the announcement. Figure B.2 compares five-minute bars of executed onshore spot trades (Yonhap Infomax) with the Bloomberg composite quotes used in Figure 1A. Across 784 matched five-minute stamps from December 2 through the December 5 session the two series track each other with a correlation of 0.99 and a median gap of 0.0 won. Trades were flat near 1,403 through the 22:35 bar and repriced sharply from the 22:40 bar, within about fifteen minutes of the start of the address.

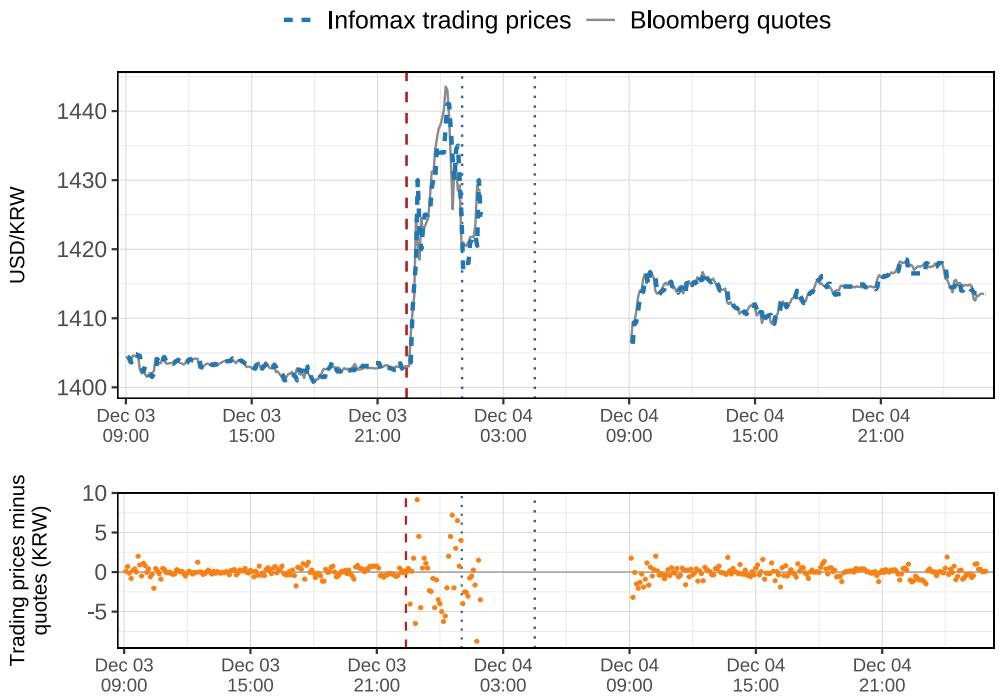
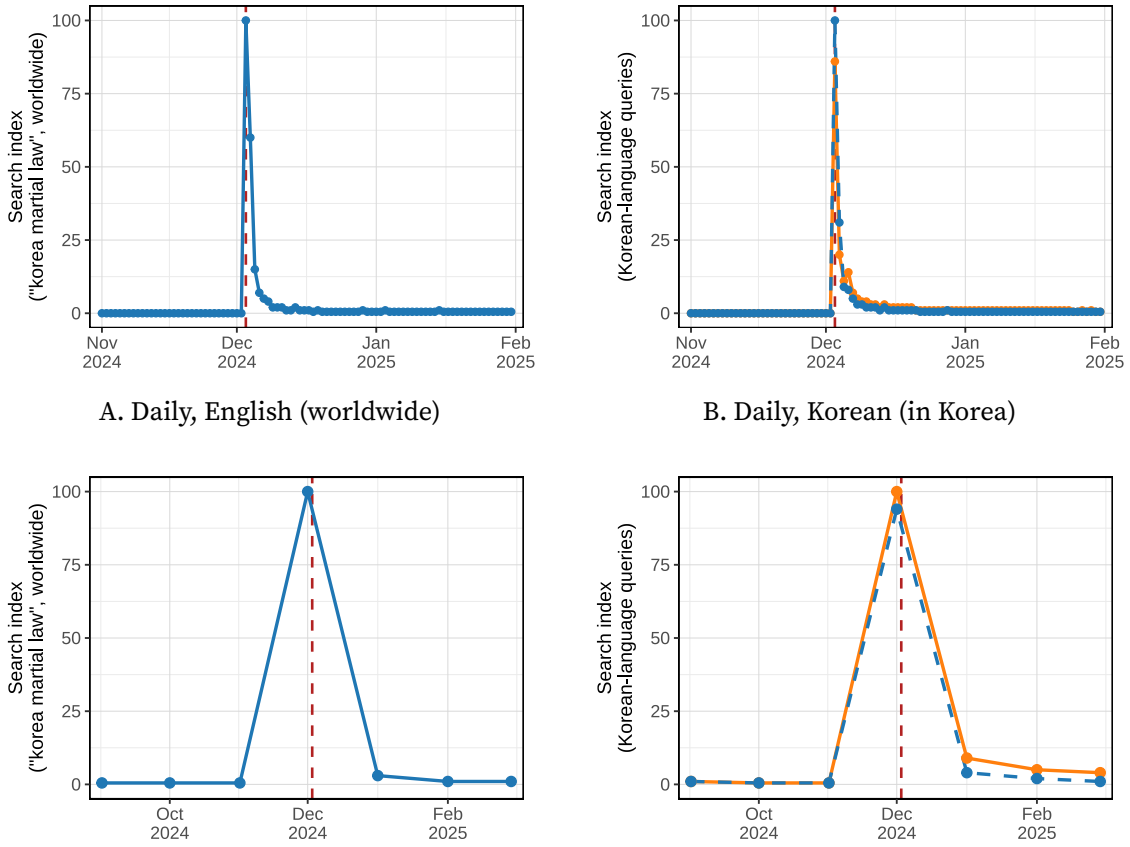


FIGURE B.2. Onshore Trading Prices and Bloomberg Quotes

Notes: Top panel: USD/KRW, December 3, 09:00–December 5, 02:00 (KST). Blue (dotted): five-minute trading prices of executed onshore spot trades in the Seoul interbank market (Yonhap Infomax, closing price of each bar). Gray: Bloomberg 5-minute quotes. Gaps are market closures, and the extended onshore session runs to 02:00. Bottom panel: the difference between the two series at matched five-minute stamps. The dashed red line marks the martial law announcement (22:23, December 3) and the dotted blue line the National Assembly vote (01:02, December 4), and the dotted gray line the withdrawal of martial law (04:30, December 4).

Google Trends. Figure B.3 shows Google search interest at daily and monthly frequency for two sets of queries. The first is the English query "Korea martial law" worldwide. The second is the Korean-language queries 계엄 (*gyeeom*, "martial law") and 계엄령 (*gyeeom-ryeong*, "martial law decree") within Korea. At daily frequency every series is exactly zero on all 32 days from November 1 through December 2 and peaks on December 3. At monthly frequency the pre-event readings never exceed 1 on the 0–100 index. These search series show no visible pre-announcement increase in attention.



A. Daily, English (worldwide)

B. Daily, Korean (in Korea)

C. Monthly, English (worldwide)

D. Monthly, Korean (in Korea)

FIGURE B.3. Google Trends by Query Language: Daily and Monthly

Notes: Google Trends relative search interest (0–100). Left column: the English query "korea martial law", worldwide. Right column: the Korean-language queries 계엄 (solid orange) and 계엄령 (dashed blue), searched within Korea. Top row: daily, November 1, 2024–January 31, 2025. Bottom row: monthly, September 2024–March 2025. Search type is Web Search, and the series were retrieved on August 15, 2026. We use the dates returned by Google Trends without conversion. The declaration at 22:23 KST on December 3 falls on December 3 under KST, UTC, and US Pacific time alike. The peak day therefore does not depend on the time-zone convention. Google scales each retrieval to 100 at its maximum, so levels are comparable within a panel but not across panels. The red dashed vertical line marks the declaration (December 3, 2024).

Appendix C. Data Sources and Sample Countries

Table C.1 summarizes the data sources used in our analysis. Table C.2 lists the 37 countries in our sample, grouped by region, along with their country codes.

TABLE C.1. Data Sources

Variable	Description	Source	Frequency
Exchange rates	Bilateral exchange rates, 37 currencies per US dollar, Jan 2000–Dec 2025	Bloomberg	Monthly
Exchange rates	Daily closing prices, 37 currencies per US dollar, Jan 2021–Apr 2025	Bloomberg	Daily
USD/KRW quote	5-minute, Dec 2–7, 2024 (KST)	Bloomberg	Intraday
USD/KRW trading price	Executed spot trades, Seoul interbank market, 5-minute, Dec 2–5, 2024 (KST)	Yonhap Infomax	Intraday
USD/KRW NDF	1-month non-deliverable forward rate, 5-minute, Dec 2–7, 2024 (KST)	Bloomberg	Intraday
TPU	Baker–Bloom–Davis Trade Policy Uncertainty index	policyuncertainty.com	Daily[a]
Google Trends	Relative search interest (0–100), English query "korea martial law" (worldwide) and Korean queries 계엄 and 계엄령 (within Korea)	Google Trends	Monthly/Daily

Notes: Exchange rate data cover 37 currencies listed in Table C.2. Daily closes are Bloomberg closing prices at each currency's own closing time. The KRW close is the 02:00 KST close of the Seoul extended session. Figure 1B displays daily closes through June 2025; the daily estimation panel ends in April 2025. [a] Daily TPU index aggregated to monthly averages for the adjustments in Appendix D.3.

TABLE C.2. Sample Countries by Region and Exchange Rate Regime

Region	Codes
Asia & Pacific	KOR, AUS, BRN[†], CHN[†], IDN, IND, ISR, JPN, KHM[†], LAO[†], MMR[†], MYS, NZL, PHL, THA, TWN[‡], VNM[†]
Europe	EUR, CHE, CZE, DNK[†], GBR, HUN, ISL, NOR, POL, RUS, SWE, TUR[†]
Americas	BRA, CAN, CHL, COL, MEX, PER
Africa	EGY[†], ZAF

Notes: [†] Managed or pegged arrangements per the IMF AREAER 2023 de facto classification (International Monetary Fund 2024). [‡] Taiwan is not covered by the AREAER and is classified as managed.

Appendix D. Robustness of the December Estimate

This appendix reports the robustness variants of Table 3 in full: alternative factor constructions, alternative control groups, and trade policy adjustments. Table D.1 first lists every specification of Table 3 and the daily counterpart. For each it gives the factor estimation sample, the event-study sample, the reference month, and the numbers of countries and observations. All specifications are estimated from a single set of files.

TABLE D.1. Estimation Samples by Specification

	Factor model		Event study			
Specification	Sample	Currencies	Sample	Ref.	Countries	*N*
Baseline	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	37	888
Alternative factor constructions						
full-sample smoother, Korea included	Feb 2000–Dec 2025	37	Jan 2024–Dec 2025	Nov 24	37	888
pre-event factor, Korea included	Feb 2000–Nov 2024	37	Jan 2024–Dec 2025	Nov 24	37	888
principal component	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	37	888
two-factor model	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	37	888
three-factor model	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	37	888
floating factor universe	Feb 2000–Nov 2024	26	Jan 2024–Dec 2025	Nov 24	27	648
Alternative control groups						
floating controls	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	27	648
best pre-event factor fit	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	19	456
Trade policy adjustments						
TPU-adjusted outcome (three variants)[a]	Feb 2000–Nov 2024	36	Jan 2024–Dec 2025	Nov 24	37	888
Daily counterpart[b]	Jan 2021–Dec 2, 2024	36	Nov 2024–Apr 2025	Dec 2	37	4,773

Notes: One row per specification of Table 3. Factor model: the sample and the number of currencies on which the factor model parameters are estimated (the 36 controls, the 26 floating controls, or all 37 currencies where Korea is included). Event study: the panel of the saturated event study, in which every month except the reference month (Ref.) carries its own Korea indicator, so the event-time indicators follow from the sample dates. Countries is the number of currencies in the event study, including Korea. [a] The TPU sensitivities are estimated on the 298 pre-event months, February 2000–November 2024. [b] Daily panel: 129 weekdays, each except the December 2 anchor carrying its own Korea indicator (Appendix E.2).

D.1. Alternative Factor Constructions

We assess whether the results are sensitive to how the global dollar factor is extracted from the data. Table D.2 compares the December estimate against six alternatives: the full-sample two-sided smoother including Korea, a pre-event factor including Korea, a pre-event principal component, two- and three-factor models, and a factor estimated on the floating currencies only. For each construction the table records the design ingredients—pre-event estimation, one-sided filtering, and the exclusion of Korea—and the placebo rank. The estimates range from 2.30 to 2.37 pp.[10]

[10]The estimate is also insensitive to the number of factors. In the pre-event control panel the first principal component accounts for 39% of the variance of exchange rate changes, the second for 8%, and the third for

The full-sample smoother uses the entire sample and is more efficient, but it lets post-event information into the outcome. The baseline forgoes that efficiency, and the December estimate is essentially unchanged either way (2.35 against 2.30 pp).

TABLE D.2. December 2024 Estimate Across Factor Constructions

Specification	Factor	Pre-event	One-sided	Korea excl.	Est. (pp)	Rank, p
Baseline	DFM, 1	Yes	Yes	Yes	2.35	3/37, 0.081
Full-sample smoother, Korea included	DFM, 1	No	No	No	2.30	3/37, 0.081
Pre-event factor, Korea included	DFM, 1	Yes	Yes	No	2.34	3/37, 0.081
Principal component	PCA, 1	Yes	—[a]	Yes	2.31	4/37, 0.108
Two-factor model	DFM, 2	Yes	Yes	Yes	2.37	3/37, 0.081
Three-factor model	DFM, 3	Yes	Yes	Yes	2.36	3/37, 0.081
Floating factor universe[b]	DFM, 1	Yes	Yes	Yes	2.32	3/27, 0.111

Notes: Saturated event study on each factor construction. Pre-event: parameters estimated on data through November 2024 only. One-sided: event-period factor from the one-sided filter. Korea excl.: Korea excluded from the factor estimation. Rank is the position of Korea's estimate among the possible treatment assignments in absolute value, re-running the row's own construction for every assignment; p is the two-sided inclusive placebo probability. [a] The principal component is static and projected with fixed pre-event weights, so no filtering applies. [b] Factor estimated on the 26 floating controls only. Rank among the 27 floating assignments.

6%. The information criteria of Bai and Ng (2002) reach their minimum at three factors but differ by 0.06 or less across one to four. Two and three factors give 2.37 and 2.36 pp with the same rank as the baseline (Table D.2). We use one factor as the parsimonious specification with a direct interpretation as the global dollar factor.

D.2. Control Group Sensitivity

Table D.3 reports two variants. Restricting the second stage to the floating controls per the IMF AREAER 2023 de facto classification (International Monetary Fund 2024) gives 2.34 pp. Restricting to the 18 controls with the best pre-event factor fit gives 2.12 pp. This group is the half of the controls with the highest pre-event R^2 of the factor model (Table H.1).

TABLE D.3. December 2024 Estimate: Control Group Sensitivity

Control group	Estimate (pp)	Rank	Placebo p
Floating controls	2.34	3 of 27	0.111
Best pre-event factor fit	2.12	4 of 37	0.108

Notes: Both groups are defined in the text. Floating controls: 648 observations, clustered standard error 0.28. Best pre-event factor fit: 456 observations, clustered standard error 0.23. Rank and inclusive two-sided placebo p-value re-run the full procedure for every possible treatment assignment. Clustered standard errors are descriptive.

D.3. Trade Policy Uncertainty

Trade policy uncertainty is a potential confounder because Korea is highly exposed to US trade policy. Figure D.1 plots the monthly Baker–Bloom–Davis trade policy uncertainty (TPU) index (Baker, Bloom, and Davis 2016). The index was already elevated in November 2024, after the US presidential election (1,291), and fell in December 2024 (1,083). The December estimate is therefore identified against a month of declining trade policy pressure. It then rose to 2,648 in March and 3,379 in April 2025 with the Liberation Day announcement.[11] Both readings are far above the 2000–2023 average of 197 and the previous peak of 1,440 in April 2020.

To check that trade policy uncertainty does not drive the December estimate, we adjust the outcome directly.[12] For each currency we regress the local factor on the TPU index over the 298 pre-event months, subtract the fitted component, and re-run the event study. Table D.4 reports the December estimate using the level, the change, or the AR(1) innovation of the index. The estimate moves by at most 0.11 pp.

The adjustment also moves Korea's placebo rank from third to fifth. Table D.5 shows why. Korea's sensitivity relative to the control average is essentially zero. Its estimated slopes are 0.03, −0.02, and −0.01 pp per standard deviation of the level, the change, and the innovation of the index, with Newey–West t-statistics between −0.14 and 0.38. Its December

[11] On April 2, 2025 the United States announced the Liberation Day reciprocal tariffs, under which Korea faced a 25% rate.

[12] Alternatively, one could include the interaction term $\text{TPU}_t \times \text{Korea}_i$. In our saturated design, however, this control is not identified because it is spanned by the Korea-month indicators of equation (2).

estimate therefore moves by at most 0.11 pp. Under the change and innovation adjustments the currencies that overtake Korea are the Myanmar kyat and the Russian ruble. Under the level adjustment they are the Turkish lira and the Australian dollar. The local factors of the kyat, the ruble, and the lira are 3 to 20 times as volatile as the panel median (4 to 28 pp per month against 1.4 pp). The sensitivity is an OLS slope of the local factor on the index, so its sampling error grows with that volatility. December 2024 then multiplies the error. The index level was 3.5 standard deviations above its pre-event mean, and its change and innovation were 2.2 and 1.7 standard deviations below theirs. A slope that is noise in an ordinary month becomes an adjustment of up to 3.5 pp, and for the kyat a change of sign. The Australian dollar is the exception. Its volatility matches Korea's, and its estimate rises by 0.48 pp because its estimated sensitivity relative to the controls is negative and the December index level is high. The rank moves because these placebos move, not Korea.

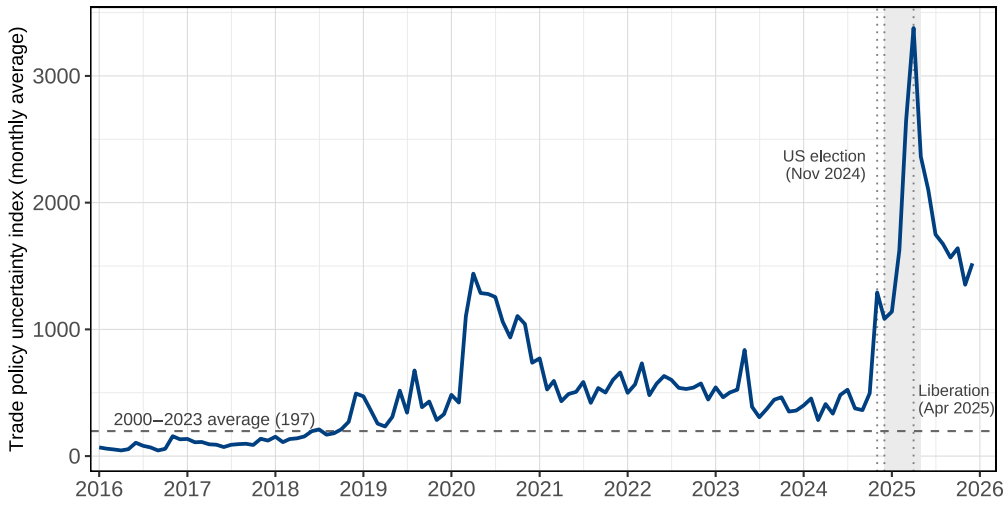


FIGURE D.1. Monthly Trade Policy Uncertainty Index, 2016–2025

Notes: Monthly average of the daily Baker–Bloom–Davis trade policy uncertainty index, January 2016–December 2025. The dashed horizontal line is the 2000–2023 average (197). Shading marks December 2024–April 2025. Dotted vertical lines mark the US election (November 2024), the martial law declaration (December 2024), and the Liberation Day tariffs (April 2025).

TABLE D.4. December 2024 Estimate Under Trade Policy Adjustments

Adjustment	Estimate (pp)	Rank of 37	Placebo p
No adjustment	2.35	3	0.081
TPU level	2.24	5	0.135
TPU change	2.31	5	0.135
TPU innovation	2.33	5	0.135

Notes: Each row regresses every currency's local factor on the Baker–Bloom–Davis trade policy uncertainty index over the 298 pre-event months, subtracts the fitted component, and re-runs the saturated event study. Rank is the position of Korea's estimate among the 37 assignments in absolute value, each assignment adjusted with its own estimated sensitivity. Placebo p is the two-sided inclusive placebo probability.

TABLE D.5. Placebo Estimates Near Korea's Rank Under the Trade Policy Adjustments

	December estimate (pp) by TPU adjustment				Local-factor
Currency	None	Level	Change	Innovation	volatility (pp)
BRL	3.82	3.01	4.60	4.45	3.35
ILS	−3.20	−3.05	−3.23	−3.23	1.53
KRW (treated)	2.35	2.24	2.31	2.33	1.72
MMK	−2.32	1.22	−3.24	−3.21	28.12
AUD	2.00	2.49	2.15	2.09	1.74
RUB	1.45	0.68	2.69	2.42	4.29
TRY	−0.85	−2.83	−1.45	−1.18	4.20
Korea's rank of 37	3	5	5	5	
Panel median volatility					1.42

Notes: The currencies listed are those ranked in the top five of the 37 treatment assignments, in absolute value, under at least one of the four outcomes. Each column re-estimates the saturated event study for every assignment with that assignment's own estimated TPU sensitivity subtracted from its outcome (Table D.4). Local-factor volatility is the pre-event standard deviation of the currency's monthly local factor, February 2000–November 2024 (Appendix H).

Appendix E. Event-Study Estimates in Full

Combining equations (1) and (2), the two-stage setup has an interactive fixed effects interpretation (Bai 2009):

$$\log E_{it} = \sum_{k \in \mathcal{K}} \beta_k \cdot (\text{Korea}_i \times \mathbf{1}[t = k]) + \alpha_i + \gamma_t + \mu_i t + \lambda_i^G F_t^G + u_{it}, \tag{E.1}$$

where F_t^G is the cumulated unobserved dollar factor with heterogeneous loadings and $\mu_i t$ is the currency's pre-event drift, which the first stage removes. The country effect α_i is a heterogeneous loading on a constant factor, and the time effect γ_t is a common factor with loadings fixed at one. We estimate the common dollar component first and then apply the event study to the residual component.

E.1. Complete Coefficients and Randomization Intervals

Table E.1 and Figure E.1 report the complete set of 23 event-time coefficients underlying Table 1 and Figure 2, with their randomization confidence intervals.[13]

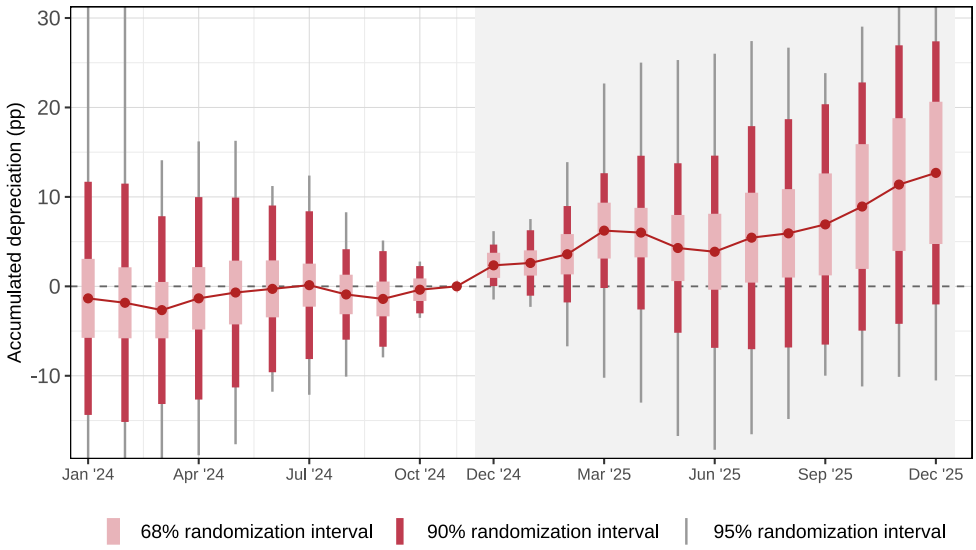


FIGURE E.1. Event-Study Estimates with Randomization Confidence Intervals

Notes: Korea's event-study coefficients from equation (2), November 2024 = 0. Intervals are constructed as in Table E.1. Shading marks the post-treatment period. Some 95% intervals extend beyond the plotted range; complete endpoints are in Table E.1.

[13]The intervals are the estimate plus or minus placebo quantiles. For each event time, the interval is the estimate plus or minus the eleventh largest, third largest, or largest absolute placebo coefficient at that event time. These give the 68, 90, and 95 percent intervals, with implied coverage of 70.3, 91.9, and 97.3 percent.

TABLE E.1. Complete Event-Time Coefficients with Randomization Ranks and Intervals

Month	Estimate (pp)	S.E.	Rank	Placebo p	68% interval	90% interval	95% interval
Jan 2024 ($k = -11$)	−1.34	1.44	26	0.703	[−5.76, 3.07]	[−14.38, 11.69]	[−39.11, 36.42]
Feb 2024 ($k = -10$)	−1.84	1.44	21	0.568	[−5.81, 2.13]	[−15.16, 11.48]	[−40.66, 36.98]
Mar 2024 ($k = -9$)	−2.66	0.92	14	0.378	[−5.81, 0.49]	[−13.16, 7.84]	[−19.41, 14.10]
Apr 2024 ($k = -8$)	−1.34	0.90	20	0.541	[−4.83, 2.15]	[−12.66, 9.98]	[−18.89, 16.20]
May 2024 ($k = -7$)	−0.69	0.87	28	0.757	[−4.26, 2.87]	[−11.30, 9.92]	[−17.65, 16.27]
Jun 2024 ($k = -6$)	−0.28	0.70	34	0.919	[−3.46, 2.90]	[−9.60, 9.04]	[−11.78, 11.22]
Jul 2024 ($k = -5$)	0.13	0.66	37	1.000	[−2.28, 2.54]	[−8.13, 8.39]	[−12.12, 12.39]
Aug 2024 ($k = -4$)	−0.91	0.47	25	0.676	[−3.12, 1.30]	[−5.96, 4.15]	[−10.10, 8.28]
Sep 2024 ($k = -3$)	−1.41	0.40	14	0.378	[−3.35, 0.54]	[−6.75, 3.94]	[−7.94, 5.13]
Oct 2024 ($k = -2$)	−0.38	0.22	28	0.757	[−1.64, 0.88]	[−3.02, 2.26]	[−3.52, 2.77]
Nov 2024 ($k = -1$, ref.)	0	—	—	—	—	—	—
Dec 2024 ($k = 0$)	2.35	0.22	3	0.081	[0.94, 3.75]	[0.03, 4.67]	[−1.47, 6.17]
Jan 2025 ($k = 1$)	2.61	0.30	6	0.162	[1.20, 4.02]	[−1.04, 6.27]	[−2.30, 7.52]
Feb 2025 ($k = 2$)	3.59	0.50	6	0.162	[1.34, 5.84]	[−1.80, 8.98]	[−6.71, 13.89]
Mar 2025 ($k = 3$)	6.23	0.69	4	0.108	[3.11, 9.35]	[−0.19, 12.65]	[−10.22, 22.68]
Apr 2025 ($k = 4$)	6.01	0.83	9	0.243	[3.24, 8.78]	[−2.59, 14.60]	[−13.00, 25.01]
May 2025 ($k = 5$)	4.29	0.88	9	0.243	[0.59, 7.98]	[−5.19, 13.76]	[−16.73, 25.30]
Jun 2025 ($k = 6$)	3.87	0.96	12	0.324	[−0.38, 8.12]	[−6.88, 14.61]	[−18.27, 26.01]
Jul 2025 ($k = 7$)	5.44	1.04	11	0.297	[0.42, 10.47]	[−7.03, 17.91]	[−16.54, 27.42]
Aug 2025 ($k = 8$)	5.93	1.06	9	0.243	[0.98, 10.87]	[−6.84, 18.69]	[−14.83, 26.68]
Sep 2025 ($k = 9$)	6.92	1.09	9	0.243	[1.22, 12.62]	[−6.51, 20.36]	[−9.99, 23.84]
Oct 2025 ($k = 10$)	8.92	1.22	8	0.216	[1.95, 15.90]	[−4.95, 22.80]	[−11.19, 29.04]
Nov 2025 ($k = 11$)	11.38	1.32	7	0.189	[3.96, 18.80]	[−4.19, 26.95]	[−10.12, 32.88]
Dec 2025 ($k = 12$)	12.69	1.37	6	0.162	[4.73, 20.64]	[−2.02, 27.39]	[−10.52, 35.89]

Notes: Saturated event study of equation (2), November 2024 reference. Clustered standard errors (S.E.) are descriptive. Rank is the position of Korea's coefficient among the 37 assignments in absolute value, and placebo p is the two-sided inclusive placebo probability. The intervals are the estimate plus or minus the eleventh-largest (68%, implied coverage 70.3%), the third-largest (90%, implied 91.9%), and the largest (95%, implied 97.3%) placebo coefficient at the same event time among the 36 placebo assignments.

Table E.2 verifies that the reference month is a pure reparameterization in the saturated design. The difference between the fitted December-versus-October and November-versus-October contrasts ("Implied December vs. November") recovers the November-anchored December estimate exactly.

TABLE E.2. Reference Month Reparameterization

	pp
December vs. October contrast	2.72
November vs. October contrast	0.38
Implied December vs. November	2.35
Main December estimate	2.35

Notes: With October 2024 as the omitted month, the difference between the two fitted contrasts recovers the November-anchored December estimate exactly (identity error 1.3e-13). In a saturated event study the reference month is a pure reparameterization.

E.2. Daily Counterpart

We apply the main procedure to daily data as a counterpart to the monthly design. The daily panel consists of the weekdays on which Bloomberg reports a close for all 37 currencies. The daily factor extraction is described in Appendix H. The local factor is cumulated and anchored at the December 2 close, so the saturated daily event study reduces to Korea's anchored level minus the control average. We denote its coefficient for weekday s by $\hat{\beta}^d_s$. The daily regression is $\tilde{E}_{is} = \alpha_i + \gamma_s + \sum_{s' \neq \text{Dec 2}} \beta^d_{s'} \cdot (\text{Korea}_i \times \mathbf{1}[s = s']) + \varepsilon_{is}$, where $\tilde{E}_{is}$ is the cumulated daily local factor anchored at the December 2 close. As in the monthly case, $\hat{\beta}^d_s$ equals Korea's anchored level minus the control average on day s. The randomization inference re-estimates the full procedure for each of the 37 currencies. The daily statistics are day by day: the one-day move $\Delta\hat{\beta}^d_s = \hat{\beta}^d_s - \hat{\beta}^d_{s-1}$ is Korea's local-factor change minus the controls' average on day s, and the anchor cancels in differencing.

Table E.3 reports Korea's daily path for every weekday of November and December 2024, with ranks and placebo p-values. Of the 3.93 pp reached by December 31, 2.33 pp accrue on the eight listed event days and 1.61 pp on the other weekdays. Korea shows no unusual pre-event divergence at daily frequency (Appendix E.3). The December average of the daily coefficients is $\bar{\hat{\beta}}^d_{\text{Dec}} = 2.22$ pp (rank 4, $p = 0.108$). The December-minus-November average, $\bar{\hat{\beta}}^d_{\text{Dec}} - \bar{\hat{\beta}}^d_{\text{Nov}} = 2.65$ pp, is the December-versus-November contrast from the daily model. The anchor at the December 2 close cancels in this difference, as it does in the one-day moves. It has the same rank and placebo p-value as the monthly estimate.

A joint randomization test on the episode levels gives the same picture as the December statistics. The mean squared level over December 5–31 ranks fourth of 37 ($p = 0.108$). The average level in excess of the level at the December 4 close, $\bar{\hat{\beta}}^{d}_{\text{Dec 5–31}} - \hat{\beta}^{d}_{\text{Dec 4}} = 1.85$ pp, ranks third ($p = 0.081$). Like every daily statistic, both are computed for each assignment from its own re-estimated path and ranked in absolute value.

E.3. Parallel Trends

Table E.1 and Figure E.1 report all ten pre-treatment coefficients. The estimates for January through October 2024 fluctuate between −2.7 and +0.1 pp with no approach path into December, and every pre-treatment randomization interval contains zero. The joint test aggregates the ten pre-treatment coefficients into one statistic per assignment and ranks Korea among the 37. On the quadratic form $\hat{\beta}'_{\text{pre}} \hat{V}^{-1} \hat{\beta}_{\text{pre}}$, the randomization analogue of a joint Wald statistic, Korea ranks 32nd of 37 ($p = 0.87$). On the largest absolute pre-treatment t-statistic it ranks 27th of 37 ($p = 0.73$). Its pre-treatment path is among the smoothest in the sample. Table E.4 collects these monthly statistics.

At daily frequency, Korea's path before the declaration shows no pre-event divergence either (Table E.5). Its average level over November, relative to the December 2 close, ranks 21st of 37 ($p = 0.57$). Its largest absolute November level ranks 16th ($p = 0.43$), and its largest absolute level over the 20 weekdays before the declaration 26th ($p = 0.70$).

TABLE E.3. Daily Excess Depreciation of the Won by Weekday, November–December 2024

November 2024							December 2024						
	Level $\hat{\beta}_s^d$			One-day move $\Delta\hat{\beta}_s^d$				Level $\hat{\beta}_s^d$			One-day move $\Delta\hat{\beta}_s^d$		
Day	pp	Rank	Placebo p	pp	Rank	Placebo p	Day	pp	Rank	Placebo p	pp	Rank	Placebo p
Nov 1	−1.54	11	0.297	—	—	—	Dec 2	0	—	—	0.45	5	0.135
Nov 4	−1.60	11	0.297	−0.06	25	0.676	Dec 3[e]	1.67	1	0.027	1.67	1	0.027
Nov 5	−1.11	18	0.486	0.49	4	0.108	Dec 4[e]	0.61	8	0.216	−1.06	1	0.027
Nov 6	0.18	33	0.892	1.28	6	0.162	Dec 5	1.11	4	0.108	0.50	5	0.135
Nov 7	−0.48	26	0.703	−0.66	9	0.243	Dec 6	1.48	6	0.162	0.37	11	0.297
Nov 8	0.11	33	0.892	0.59	9	0.243	Dec 9[e]	2.14	2	0.054	0.66	7	0.189
Nov 11	0.16	34	0.919	0.05	33	0.892	Dec 10	2.14	2	0.054	−0.00	37	1.000
Nov 12	0.38	26	0.703	0.22	25	0.676	Dec 11	1.85	7	0.189	−0.29	12	0.324
Nov 13	0.01	37	1.000	−0.37	9	0.243	Dec 12[e]	1.79	6	0.162	−0.06	24	0.649
Nov 14	−0.13	30	0.811	−0.14	24	0.649	Dec 13	2.03	4	0.108	0.24	22	0.595
Nov 15	−0.43	19	0.514	−0.30	11	0.297	Dec 16[e]	2.21	4	0.108	0.18	16	0.432
Nov 18	−0.51	17	0.459	−0.08	25	0.676	Dec 17	1.94	6	0.162	−0.27	13	0.351
Nov 19	−0.64	12	0.324	−0.13	21	0.568	Dec 18	1.72	9	0.243	−0.22	16	0.432
Nov 20	−0.25	26	0.703	0.39	2	0.054	Dec 19	2.21	7	0.189	0.49	11	0.297
Nov 21	−0.13	31	0.838	0.12	28	0.757	Dec 20	2.21	6	0.162	0.00	37	1.000
Nov 22	0.08	33	0.892	0.22	17	0.459	Dec 23	2.62	4	0.108	0.41	10	0.270
Nov 25	−0.12	30	0.811	−0.20	23	0.622	Dec 24	3.12	4	0.108	0.50	6	0.162
Nov 26	−0.57	11	0.297	−0.46	5	0.135	Dec 25	3.01	4	0.108	−0.11	8	0.216
Nov 27	−1.15	5	0.135	−0.57	8	0.216	Dec 26[e]	3.70	4	0.108	0.69	3	0.081
Nov 28	−0.70	5	0.135	0.44	5	0.135	Dec 27[e]	3.72	2	0.054	0.01	36	0.973
Nov 29	−0.45	5	0.135	0.26	8	0.216	Dec 30	3.70	1	0.027	−0.01	37	1.000
							Dec 31[e]	3.93	2	0.054	0.23	11	0.297

Statistics combining several weekdays	pp	Rank	Placebo p
Average level, December ($\bar{\hat{\beta}}^d_{\text{Dec}}$)	2.22	4	0.108
December minus November average ($\bar{\hat{\beta}}^d_{\text{Dec}} - \bar{\hat{\beta}}^d_{\text{Nov}}$)	2.65	3	0.081
Average level, December 5–31, minus the December 4 level	1.85	3	0.081
Mean squared level, December 5–31 (pp^2)	6.69	4	0.108

Notes: Korea's daily event-study coefficient $\hat{\beta}_s^d$ is the cumulated local factor anchored at the December 2 close minus the control average, and $\Delta\hat{\beta}_s^d$ is its change from the previous weekday (Appendix E.2). Rank is the position of Korea's statistic among the 37 assignments in absolute value, and placebo p is the two-sided inclusive placebo probability. December 2 is the anchor. [e] Event days of Table 2. The bottom panel collects the December statistics that combine several weekdays (the November pre-event statistics are in Table E.5), each computed for every assignment from its own path and ranked in absolute value; the mean squared level is ranked on its value. The panel covers the weekdays on which Bloomberg reports a close for all 37 currencies. The move on November 1 is omitted because the table starts on that day.

TABLE E.4. Monthly Pre-Event Statistics for Korea

	Korea	Rank of 37	Placebo p
October 2024 coefficient, $k = -2$ (pp)	−0.38	28	0.757
Joint quadratic form, January–October 2024	74.9	32	0.865
Largest absolute pre-treatment t-statistic	3.48	27	0.730

Notes: Pre-treatment coefficients are the ten November-anchored event-study coefficients for January–October 2024 (Table E.1). The quadratic form is $\hat{\beta}'_{\text{pre}}\hat{V}^{-1}\hat{\beta}_{\text{pre}}$ with the clustered covariance $\hat{V}$, and the t-statistic uses the clustered standard error. Rank is the position of Korea's statistic among the 37 possible treatment assignments in absolute value, each assignment computed from its own re-estimated path. Placebo p is the inclusive two-sided placebo probability.

TABLE E.5. Daily Pre-Event Statistics for Korea

	Korea (pp)	Rank of 37	Placebo p
Average level over November 2024, $\bar{\hat{\beta}}^{d}_{\text{Nov}}$	−0.42	21	0.568
Largest absolute level in November 2024	1.60	16	0.432
Largest absolute level, 20 weekdays before December 3	1.15	26	0.703

Notes: Levels are the daily event-study coefficients $\hat{\beta}^{d}_{s}$, Korea's cumulated local factor minus the control average at daily frequency, anchored at the December 2, 2024 close (= 0), as constructed in Appendix E.2. Rank is the position of Korea's statistic among the 37 possible treatment assignments in absolute value, each assignment computed from its own re-estimated daily path. Placebo p is the inclusive two-sided placebo probability.

Appendix F. Randomization Inference and the Two Larger Placebo Estimates

For each of the 37 currencies in turn, the full two-stage procedure is re-estimated with that currency designated treated and the December event date held fixed. Every other currency, KRW included, serves as a control for each assignment. We report inclusive two-sided placebo p-values. [14] Treatment was not assigned at random, so the placebo p-values are benchmarks that treat the 37 assignments as exchangeable rather than exact randomization probabilities.

Korea's December coefficient ranks third of 37 in absolute value, which is a placebo p-value of 0.081 (Table 1).[15] Of the two placebo estimates that exceed Korea's in absolute value, only one is a depreciation.

The largest placebo estimate, 3.82 pp, belongs to Brazil. The real fell to a record low in December 2024 after the government's fiscal package of late November disappointed mar-

[14] One could also remove KRW from the control pool to avoid contamination from the actual event. Doing so leaves Korea's December rank and p-value unchanged and moves no other rank in Tables E.1 and G.1 by more than one position.

[15] Studentizing leaves Korea's December rank unchanged. The clustered standard errors are similar across assignments, between 0.20 and 0.23 pp, because the treated currency's residuals are zero in the saturated design and the standard error depends only on the dispersion of the control outcomes.

kets. The Banco Central do Brasil intervened repeatedly, selling about US$21.6 billion in spot auctions during the month, with further dollar repurchase operations (International Monetary Fund 2025). At monthly frequency the real's December depreciation, 5.2 pp, exceeds the won's 3.2 pp (Figure F.1, left panel). The second placebo estimate, −3.20 pp, belongs to Israel, whose currency strengthened toward year-end amid the ceasefire with Hezbollah and a decline in Israel's risk premium (Bank of Israel 2025, Chapter 1). Both movements are country-specific episodes unrelated to Korea. Both currencies also have far less parallel pre-treatment paths than Korea. Their joint quadratic forms are 626 and 3,541 against Korea's 75, and their largest absolute pre-treatment t-statistics are 12.1 and 12.9 against 3.5 (Appendix E.3). At daily frequency the picture is sharper (Figure F.1, right panel): Brazil's depreciation built up over the second half of December, Israel moved in the opposite direction, and neither shows the won's December 3 jump. Figure F.2 plots the

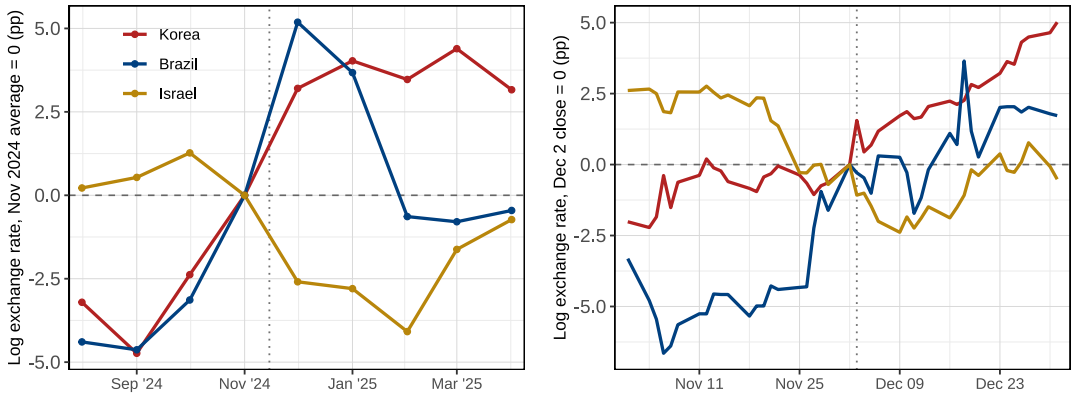


FIGURE F.1. Korea, Brazil, and Israel: Exchange Rates at Monthly and Daily Frequency

Notes: Log USD exchange rates in percentage points, an increase denoting depreciation against the US dollar. Left: monthly averages, August 2024–April 2025, November 2024 = 0, the dotted line marking the December onset. Right: daily closes, November 1–December 31, 2024, December 2 close = 0, the dotted line marking the declaration.

complete coefficient paths of all 37 assignments through December 2025, with Brazil and Israel highlighted (an extended version of Figure 2). Korea's path lies inside the placebo distribution throughout the pre-treatment period and moves to the third-largest position in absolute value in December.

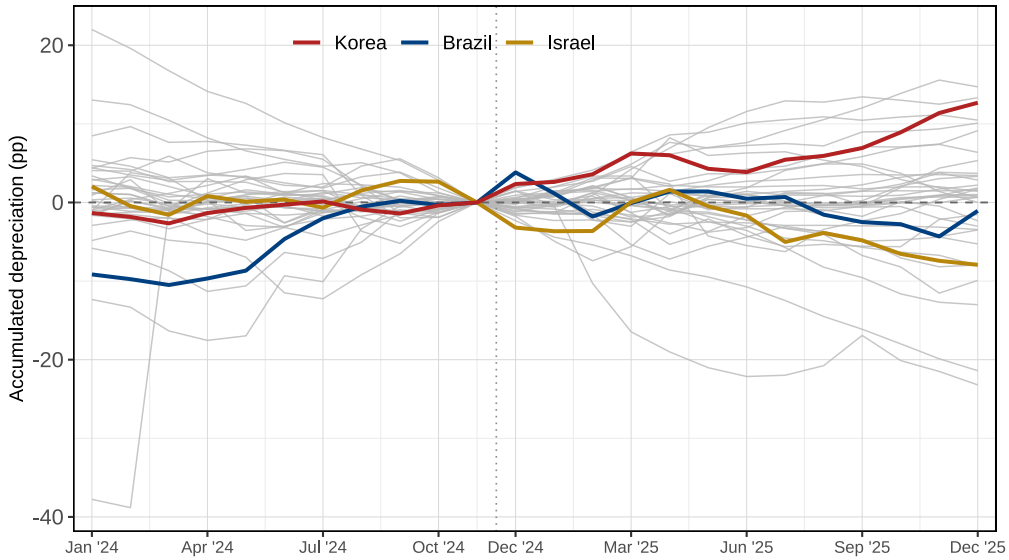


FIGURE F.2. Placebo Coefficient Paths, January 2024–December 2025

Notes: Same construction as Figure 2, full window, with Brazil and Israel highlighted.

Appendix G. The Path After December 2024

Table G.1 reports the randomization tests of the path after December 2024 through December 2025. The results show that no month after December stands out from the placebo distribution. The month-by-month p-values against zero are 0.108 or larger. Neither is any month distinguishable from the December estimate, with equality p-values of 0.162 or larger.

TABLE G.1. Randomization Tests of the Path After December 2024

	Estimate	p vs. zero	Diff. from Dec	p vs. Dec
Jan 2025	2.61	0.162	0.26	0.838
Feb 2025	3.59	0.162	1.24	0.568
Mar 2025	6.23	0.108	3.88	0.243
Apr 2025	6.01	0.243	3.66	0.297
May 2025	4.29	0.243	1.94	0.514
Jun 2025	3.87	0.324	1.52	0.622
Jul 2025	5.44	0.297	3.09	0.405
Aug 2025	5.93	0.243	3.58	0.405
Sep 2025	6.92	0.243	4.58	0.351
Oct 2025	8.92	0.216	6.57	0.270
Nov 2025	11.38	0.189	9.03	0.216
Dec 2025	12.69	0.162	10.34	0.162

Notes: Estimates in pp. Both p-values are two-sided inclusive placebo probabilities from the same 37-assignment randomization as the December estimate in Table 1, applied to the month's coefficient and to its difference from the December coefficient. December itself has $p = 0.081$.

Appendix H. Dynamic Factor Model

Monthly. The parameters are estimated on the complete pre-event control panel, February 2000–November 2024, with no missing observations. The series are standardized with pre-event moments and the idiosyncratic covariance is diagonal. The expectation maximization algorithm of Doz, Giannone, and Reichlin (2012) is initialized by the first principal component, as in the two-step estimator of Doz, Giannone, and Reichlin (2011). The factor sign is set so that the median control loading is positive, and the estimated autoregressive coefficient is 0.36. All 36 control currencies remain in the estimation sample. Currencies with large discrete devaluations, such as the Egyptian pound and the Myanmar kyat, remain in the panel. Their estimated loadings are close to zero (Table H.1), so they carry little weight in the factor.

Table H.1 reports the estimated mean $\hat{\mu}_i$, the standardized loading, and the pre-event R^2 for each currency. All 36 control loadings are positive, meaning every currency depreciates when the factor rises, and Korea's projected loading is likewise positive (0.17, with $R^2 = 0.40$). The loadings are largest for the European currencies, with R^2 up to 0.87, while several other currencies have loadings close to zero.

Figure H.1 shows the decomposition for Korea. The fitted common component in

panel (a) tracks the broad dollar movements in the won, while the December 2024 depreciation is not absorbed by the global component and appears in the local factor of panel (b), the outcome of the event study.

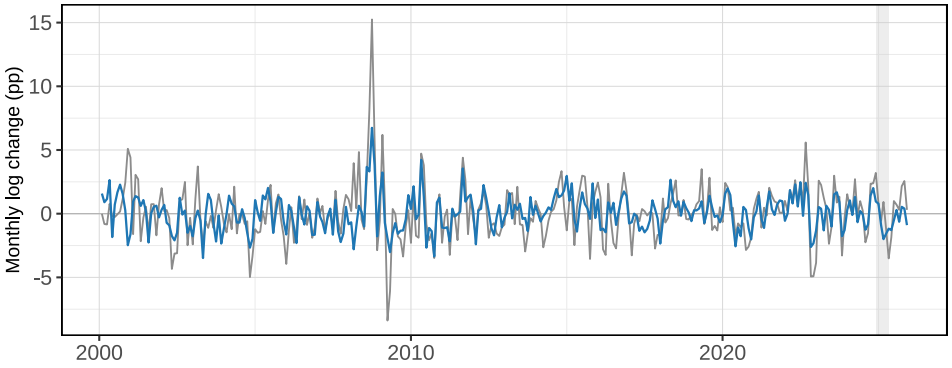
TABLE H.1. Factor Means, Loadings, and Pre-Event Fit by Currency

Currency	Mean	Loading	R^2	Currency	Mean	Loading	R^2
EUR	-0.02	0.25	0.87	CLP	0.21	0.15	0.31
DKK	-0.02	0.25	0.87	ILS	-0.03	0.14	0.30
CZK	-0.13	0.24	0.83	INR	0.22	0.13	0.24
SEK	0.08	0.24	0.82	COP	0.28	0.13	0.23
HUF	0.14	0.24	0.80	IDR	0.26	0.12	0.22
PLN	-0.00	0.23	0.74	BRL	0.39	0.12	0.22
NOK	0.11	0.22	0.72	PHP	0.12	0.11	0.18
BND	-0.08	0.22	0.67	MXN	0.25	0.11	0.18
CHF	-0.20	0.21	0.63	PEN	0.03	0.11	0.16
AUD	0.00	0.21	0.61	CNY	-0.05	0.10	0.15
NZD	-0.05	0.19	0.54	TRY	1.39	0.10	0.14
GBP	0.08	0.19	0.54	JPY	0.13	0.09	0.12
CAD	-0.01	0.17	0.43	RUB	0.43	0.08	0.10
KRW	0.07	0.17	0.40	VND	0.20	0.05	0.03
THB	-0.03	0.16	0.38	LAK	0.38	0.04	0.02
TWD	0.02	0.16	0.37	KHR	0.02	0.02	0.00
MYR	0.05	0.16	0.35	EGP	0.90	0.01	0.00
ZAR	0.36	0.15	0.32	MMK	1.95	0.01	0.00
ISK	0.22	0.15	0.31				

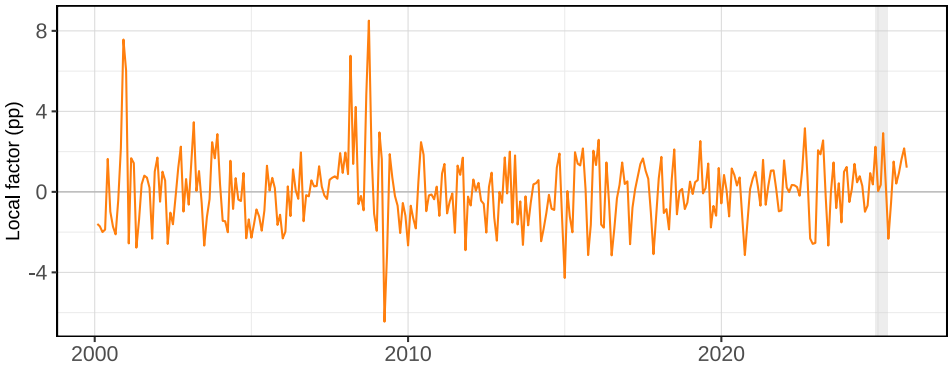
Notes: Pre-event mean monthly log change $\hat{\mu}_i$ (pp), standardized loadings, and pre-event R^2 (February 2000–November 2024). The model is estimated on the 36 control currencies. Korea's loading is obtained by projecting KRW changes on the pre-event factor. Sorted by R^2.

Daily. The daily counterpart of Appendix E.2 extracts the factor with the same architecture at daily frequency. Daily log changes of the 36 control currencies enter the DFM over the January 2021–December 2, 2024 window. The same standardization, diagonal idiosyncratic covariance, and EM estimation apply. The estimated autoregressive coefficient of the daily factor is −0.01, so the daily factor is essentially serially uncorrelated.

Table H.2 reports the daily means, loadings, and estimation fit. The factor is dominated by the European currencies, with R^2 up to 1.00 for the euro and the Danish krone, while several other currencies have loadings close to zero. Korea's projected loading is 0.05 with $R^2 = 0.04$, so the dollar factor explains little of daily KRW variation and the daily decomposition passes most of the won's movement into the local factor. Figure H.2 shows the fit for Korea. The fitted global component is small at daily frequency, and the



A. Monthly log change of USD/KRW: actual and fitted common component



B. Local factor

FIGURE H.1. Dynamic Factor Model Fit for Korea

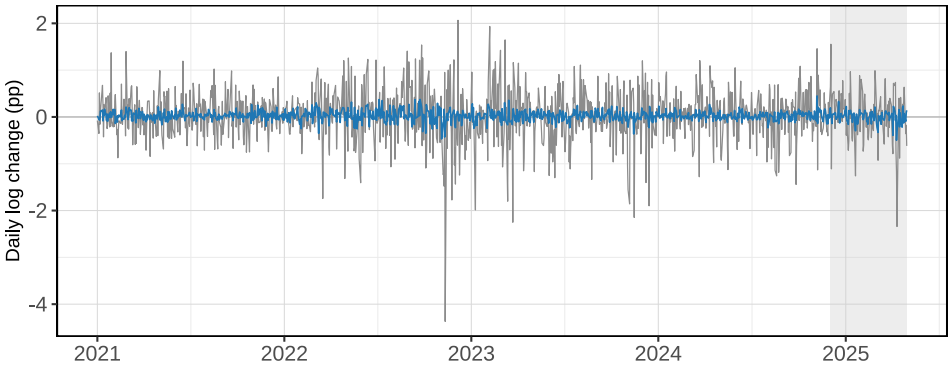
Notes: Panel (a): monthly log change of USD/KRW (gray) and the fitted common component $\hat{\mu}_{KRW} + \hat{\lambda}^{G}_{KRW}\hat{f}^{G}_{t}$ (blue). Panel (b): the local factor $\hat{e}_{KRW,t} = \Delta s_{KRW,t} - \hat{\mu}_{KRW} - \hat{\lambda}^{G}_{KRW}\hat{f}^{G}_{t}$. Parameters are estimated on the pre-event sample and the event period is filtered one-sidedly. Shading marks the political risk episode (December 2024–April 2025).

December 3 spike appears almost entirely in the local factor of panel (b).

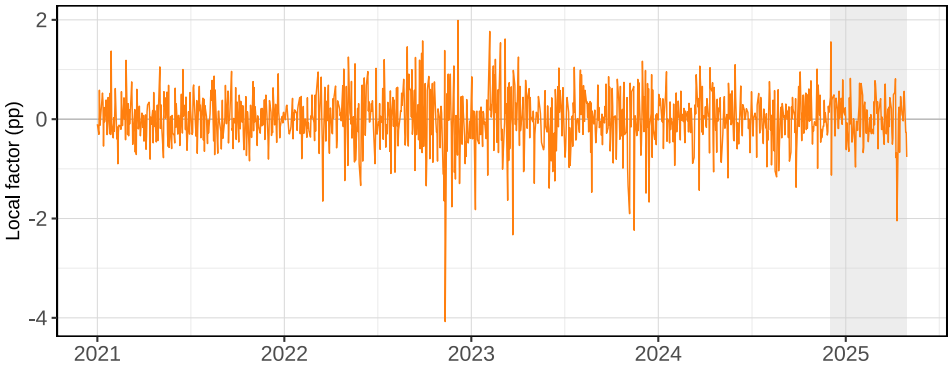
TABLE H.2. Daily Factor Means, Loadings, and Pre-Event Fit by Currency

Currency	Mean	Loading	R^2	Currency	Mean	Loading	R^2
DKK	0.015	0.28	1.00	CLP	0.031	0.11	0.15
EUR	0.015	0.28	1.00	MXN	0.002	0.10	0.14
CZK	0.011	0.24	0.77	COP	0.026	0.10	0.13
PLN	0.009	0.23	0.70	BRL	0.015	0.08	0.07
SEK	0.028	0.23	0.69	PEN	0.004	0.07	0.07
ISK	0.008	0.22	0.62	INR	0.014	0.05	0.04
GBP	0.008	0.21	0.59	KRW	0.025	0.05	0.04
HUF	0.028	0.21	0.58	MYR	0.010	0.05	0.03
CHF	0.000	0.20	0.54	TWD	0.015	0.05	0.03
NOK	0.025	0.20	0.52	IDR	0.012	0.04	0.03
BND	0.002	0.20	0.49	RUB	0.036	0.04	0.02
AUD	0.017	0.19	0.46	PHP	0.020	0.03	0.02
NZD	0.019	0.19	0.45	TRY	0.151	0.02	0.00
CAD	0.010	0.16	0.33	VND	0.009	0.01	0.00
THB	0.014	0.15	0.29	KHR	-0.000	-0.01	0.00
ZAR	0.021	0.15	0.28	MMK	0.045	-0.01	0.00
CNY	0.011	0.13	0.22	LAK	0.084	-0.00	0.00
JPY	0.036	0.12	0.20	EGP	0.113	0.00	0.00
ILS	0.013	0.12	0.19				

Notes: Pre-event mean daily log change $\hat{\mu}_i$ (pp), standardized loadings, and pre-event R^2 at daily frequency (January 2021–December 2, 2024). The model is estimated on the 36 control currencies. Korea's loading is obtained by projecting standardized KRW changes on the smoothed pre-event factor and is reported in its sorted position.



A. Daily log change of USD/KRW: actual and fitted common component



B. Local factor

FIGURE H.2. Daily Factor Model Fit for Korea

Notes: Panel (a): daily log change of USD/KRW (gray) and the fitted common component (blue), January 2021–April 2025. Panel (b): the daily local factor. Parameters are estimated on January 2021–December 2, 2024, and the event period is filtered one-sidedly. Shading marks the political risk episode (December 2024–April 2025).